Zexu Chen

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Education

The Ohio State University — PhD, Materials Science and Engineering Aug 2022 – May 2028 (expected)

Advisor: Prof. Yunzhi Wang

The Ohio State University — MS, Materials Science and Engineering (earned en route to PhD) Conferred May 2025

Sichuan University / The Ohio State University — BSc, Materials Science Aug 2017 – May 2022

Publications

- **Chen Z.**, Sriram H., Yu T., Feng L., Zhu J., Rajagopalan J., Wang Y. (2026). Programming Martensitic Transformation and Mechanical Response in NiTi Shape Memory Alloys through One-Dimensional Compositional Modulations. Under review at *Acta Materialia*; preprint SSRN 7302838. https://doi.org/10.2139/ssrn.7302838
- Sriram H., Feng L., **Chen Z.**, Wang Y. (2026). Soft and Hard Confinement Effects on Martensitic Transformation in NiTi Shape Memory Alloys: Insights from Phase-Field Simulations. *Shape Memory and Superelasticity.* https://doi.org/10.1007/s40830-026-00619-3
- Abe O.O., Qiu Z., **Chen Z.**, Jinschek J.R., Gouma P.-I. (2021). Effect of Crystallite Size on the Low-Temperature Solid-Solid Phase Transformations in the WO_3 System. *Ceramics International* 47, 33476–33482. https://doi.org/10.1016/j.ceramint.2021.08.254

Research Experience

Graduate Researcher, Wang Group, The Ohio State University (NSF DMR-2333551) *Aug 2022 – Present*

- Apply Ni_4Ti_3 precipitation field to GPU-accelerated 3D phase-field framework (Allen-Cahn / Cahn-Hilliard, OpenACC, SLURM on OSC) for martensitic transformation in NiTi alloys.
- Demonstrated linear superelastic response via concentration modulation and amorphous-crystalline composite microstructures; published in *Shape Memory and Superelasticity*.
- *Ongoing (Jul 2025 –):* ML-accelerated inverse design of compositionally graded SMAs with Profs. T. Zhang (HKUST) and J. Rajagopalan (ASU), using GPR, LASSO and Bayesian-optimization surrogates on CALPHAD-informed data.
- *Ongoing (Nov 2025 –):* multiscale modeling of shear band control in layered metallic glass/crystalline composites (phase-field + STZ kinetic Monte Carlo) with Prof. Y. Fan (U. Michigan); NSF CMMI-2611448.
- *Ongoing (May 2026 –):* phase-field modeling of grain boundary concentration modulations in polycrystalline NiTi.

ML for Superconductivity Prediction, MATSCEN 7193.02 (Machine Learning for Materials Research) *Spring 2026*

- Built XGBoost and GPR pipeline to predict superconducting parameters (λ, T_c) from Eliashberg spectral functions $\alpha^2F(\omega)$, using physics-informed features (frequency-weighted α^2F/ω bins) aligned with the Eliashberg integral.
- Tuned GPR with Matérn ($\nu = 1.5/2.5$) + White kernels to capture spectral discontinuities, and benchmarked bin resolution (up to 800) vs. generalization to characterize the curse-of-dimensionality trade-off.

Teaching

- Graduate Teaching Assistant, OSU (Oct 2023 – Oct 2024): *Introduction to Engineering Materials*; *Modeling and Simulation Lab II*.

Conference Presentations

- TMS 2027 (3 orals, accepted), TMS 2026 (poster), MS&T 2025 (oral), TMS 2025 (poster), MS&T 2023 (oral) — phase-field modeling of structural/compositional inhomogeneous NiTi shape memory alloys.

Professional Service

- Reviewer, *Shape Memory and Superelasticity* (ASM International / Springer), 2026 – Present.
- Reviewer, *Metallurgical and Materials Transactions A* (TMS / ASM International / Springer), 2026 – Present.

Technical Skills

Programming: Python (NumPy, SciPy, scikit-learn, PyTorch), C, MATLAB, Bash
HPC / GPU: GPU acceleration (CUDA, OpenACC), SLURM, Linux, Git, Ohio Supercomputer Center
Simulation: Phase-field (Allen-Cahn, Cahn-Hilliard, multiphase-field), Khachaturyan microelasticity, STZ kinetic Monte Carlo, CALPHAD (Thermo-Calc), LAMMPS, COMSOL, OVITO, ParaView
Machine Learning: Gaussian Process Regression, Bayesian optimization, XGBoost, LASSO, CNNs, generative models (cVAE, VAE-GAN), Physics-Informed Neural Networks
Domain: Shape memory alloys (NiTi), martensitic transformation, ICME, materials informatics
Languages: English (fluent), Mandarin (native)